# Lesson 10.1 - Hypothesis Testing - Null and Alternative Hypotheses

> **Video:** Hypothesis Testing - Null and Alternative Hypotheses by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

## The two hypotheses

| | Null hypothesis $H_0$ | Alternative hypothesis $H_a$ (or $H_1$) |
|---|---|---|
| What it is | The **status quo**: the currently accepted value or claim | The **challenger**: what the researcher is trying to show |
| Sign | Contains the **equality**: $=, \leq, \geq$ | **Strict** inequality: $\neq, <, >$ |

The researcher tries to gather enough evidence to **reject** $H_0$ in favor of $H_a$.

> **Careful wording:** if the evidence isn't strong enough, we **fail to reject** $H_0$. That doesn't prove $H_0$ is true, or even that it is "likely" true. It only means the data didn't rule it out.

## Steps

1. **Mean or proportion?**
   - "average", "mean" → population mean $\mu$
   - "percent", "proportion", "fraction" → population proportion $p$, written as a decimal (80% → 0.80)
2. **Find the accepted value** (the status quo) and write $H_0$ with the equality.
3. **Find what the researcher suspects** and write $H_a$ with the strict inequality. Together, $H_0$ and $H_a$ cover every possibility.

| Wording | $H_0$ | $H_a$ |
|---|---|---|
| "is different from", "is not" | $=$ | $\neq$ |
| "at least", "no less than" | $\geq$ | $<$ |
| "at most", "no more than" | $\leq$ | $>$ |

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## Examples

**1. Calculator mass.** Company XYZ's calculators average **450 g**. An engineer thinks the average is different and weighs 50 calculators.

$$H_0 : \mu = 450 \qquad H_a : \mu \neq 450$$

("Average" means a mean.)

**2. Graduation rate.** Teachers believe **at least 80%** of students graduate. A student disagrees.

$$H_0 : p \geq 0.80 \qquad H_a : p < 0.80$$

("Percent" means a proportion. "At least 80%" means 80% or more.)

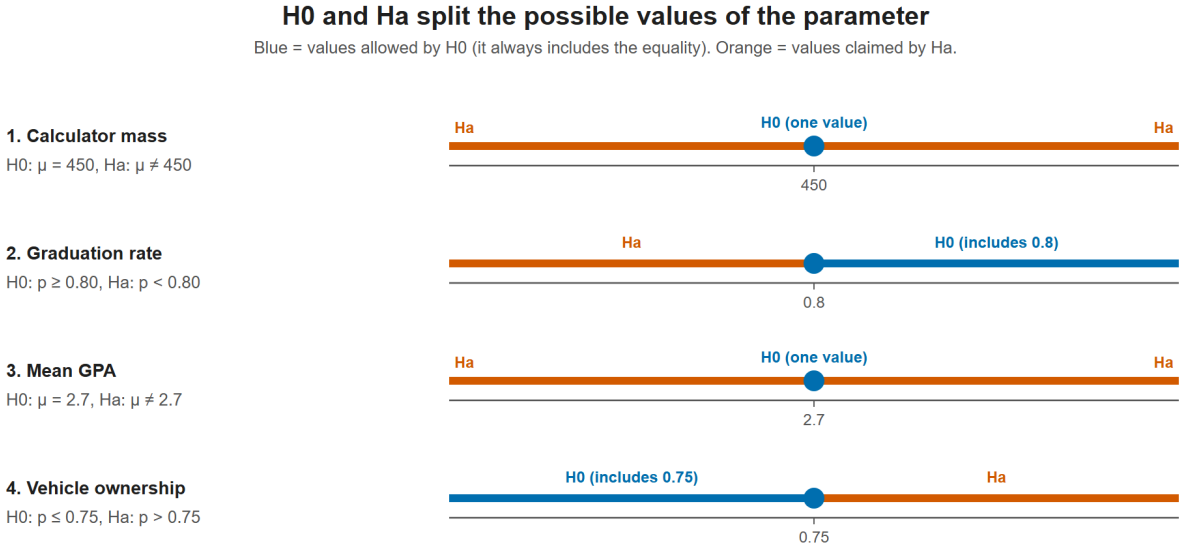
**3. GPA.** A teacher wants to test whether the mean GPA is **different from 2.7**. Here it is easier to start with $H_a$, the teacher's claim.

$$H_0 : \mu = 2.7 \qquad H_a : \mu \neq 2.7$$

**4. Vehicle ownership.** The share of residents who own a vehicle is **no more than 75%**. A researcher disagrees and surveys 100 residents.

$$H_0 : p \leq 0.75 \qquad H_a : p > 0.75$$

("No more than" means ≤. Surveying 100 residents gives the sample size, which is not part of either hypothesis.)




H0 and Ha always cover every possible value between them. H0 holds the boundary value (the equality). With "≠", H0 is a single point and Ha is everything else.

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## Key takeaways

- $H_0$ = status quo with the equality ($=$, $\leq$, $\geq$). $H_a$ = the claim being tested, with a strict sign ($\neq$, $<$, $>$).
- Hypotheses are about **population parameters** ($\mu$ or $p$), never about sample values like $\bar{x}$ or $\hat{p}$.
- Translate the wording carefully: "at least" → $\geq$, "no more than" → $\leq$, "different" → $\neq$.